Quriosity

LE HUB QUANTIQUE

Fault-tolerant quantum computation: from resource estimations to experimental realization

Paul Hilaire • Inria / Télécom Paris • 22 October 2026

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Inria

About me

Paul Hilaire
QURIOSITY (Inria / Télécom Paris)
Chaire de Professeur Junior

- Experimental PhD in Quantum Optics (2019)
- Postdoc in MBQC and quantum repeaters (2019-2022)
- Lead Quantum Information Scientist at Quandela (2022-2025)

- **Quantum error correction**
- **Fault-tolerant architectures**
- **Photonic quantum computing**

Teach QEC and Quantum Hardware at M2 QMI
(master in mathematical aspects of quantum information)

Ambitions of this talk

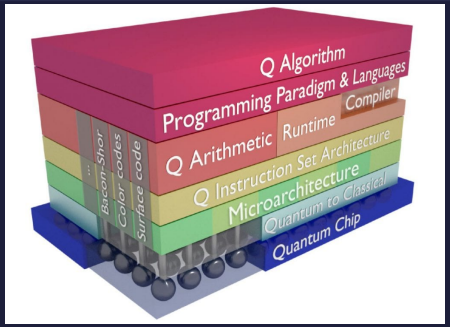


Figure: Stacks of a FT Quantum Computer

credit: QuTech

- Motivate fault-tolerance in quantum computing
- Introduce quantum error correction
- Challenges of building a fault-tolerant quantum computer
- When will fault-tolerance be achieved?
  - Experimental progress
  - Resource estimation

Ambitions of this talk

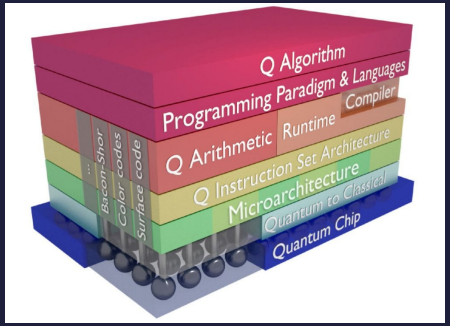


Figure: Stacks of a FT Quantum Computer

- Global overview of where FTQC is at
- Not too technical
- But not to high level as well
- Scientifically honest
- Not a company roadmap

credit: QuTech

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Quantum Computing objectives

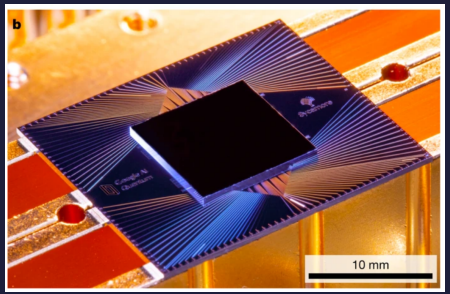


Figure: Google's Sycamore chip: Quantum advantage experiment

- Quantum advantage (supremacy) . . .
- . . . for useful applications.
- The challenge is noise.
- Objective: Building Noisy-Intermediate Scale Quantum (NISQ) devices for near-term applications.

Arute et al., Nature 574, 505–510 (2019)

A closer look at errors

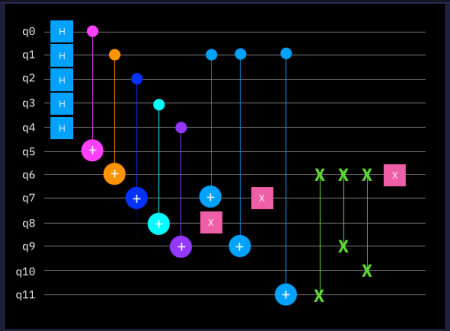


Figure: Quantum circuit model

- Q. Algo: sequence of operations on qubits.
- n_G noisy operations, error rate p each.
- Intuition: Algo. success probability $\propto P_{\text{no error}}$.

credit: Qiskit

Back of the enveloppe calculation



$$P_{\text{no error}} \approx (1-p)^{n_G}$$

$$p \ll 1$$

$$P_{\text{no error}} \approx 1 - n_G p$$

Algorithm success requires

$$n_G = O(1/p)$$

A closer look at errors

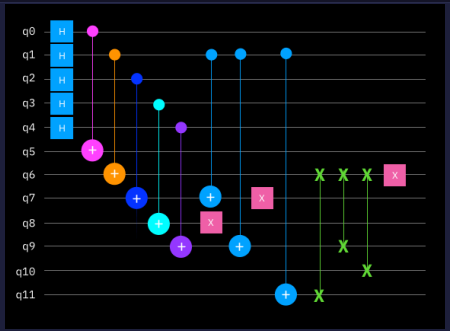


Figure: Quantum circuit model

- Q. Algo: sequence of operations on qubits.
- n_G noisy operations, error rate p each.
- Intuition: Algo. success requires $n_G = O(1/p)$
- Objectives:
  - Reduce n_G (NISQ)
  - Reduce p (hardware)
  - Replace p with p_L (FTQC)

credit: Qiskit

NISQ solution

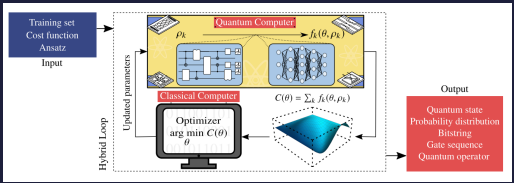


Figure: Variational Quantum Algorithm (VQA) for NISQ devices

- Main challenge: finding short-depth circuits for useful problems.
- Second challenge: no complexity proofs of advantage.
- Candidates: variational algorithms, quantum simulation.
- So far no success.
- Error handling: Hardware improvements and error mitigation.

Cerezo et al., Nat. Rev. Phys. 3, 625-644 (2021)

Applications of quantum computing



Application	p_L	Reference
Shor's algorithm (RSA-2048)	$\sim 10^{-12}$	Gidney and Ekerå (2021), *Quantum*
FeMoco	10^{-14}–10^{-17}	Otten et al. (2023), QREChem
Other molecular chemistry	10^{-9}–10^{-17}	Otten et al. (2023), QREChem
Fermi–Hubbard model	10^{-8}–10^{-9}	Kan and Symons (2025), *npj Quantum Information*
High-T superconductors	$\sim 2 \times 10^{-9}$	Kan and Symons (2025), *npj Quantum Information*

Applications with provable advantage (and their target error rates)

NISQ: useful in niches. Scalable advantage needs QEC.

Numbers depend on algorithmic + hardware assumptions.

How to reduce the error rate? Hardware improvements

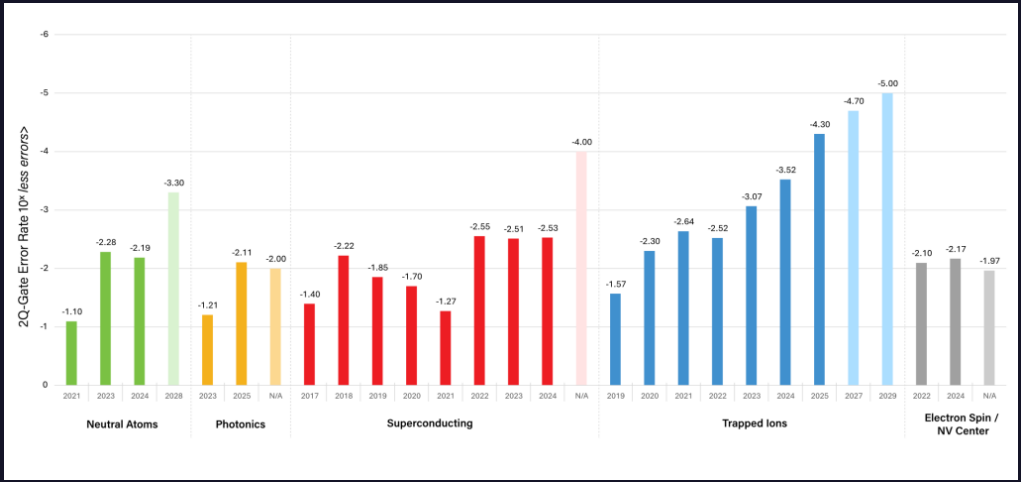


Figure: Improvement of 2Q-gate fidelity over the years, MIT Benchmarking Survey

Fault-tolerant quantum computation

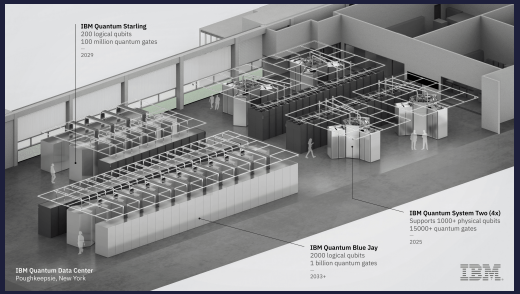


Figure: IBM's Fault-Tolerant Quantum Computer roadmap

- Physical errors cannot be reduced to arbitrarily low values.
- Other strategies to reduce errors?
- Classical computing strategies? Error correction
- FTQC idea: exploit error correction on quantum information.

How hard is it?

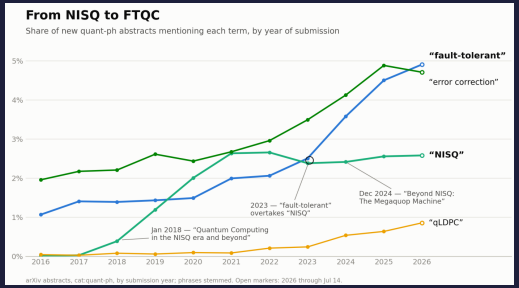


Figure: keywords in arXiv preprint submissions

- 2018: *Because of the **very hefty overhead cost** of quantum error correction, the era of fault tolerant quantum computing (FQTC) may still be rather distant. No one really knows how long it will take to get there.* — John Preskill, Quantum Computing in the NISQ era and beyond
- 2026: Preskill, founder of Oratomic, a neutral-atom FTQC start up.
- Talk's objectives:
  - What makes it hard?
  - Easier with recent developments.

Following outline

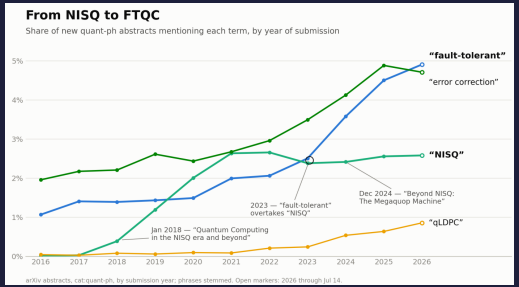


Figure: keywords in arXiv preprint submissions

- High-level view on FTQC
  - High-level view of QEC.
  - The surface code and qLDPC codes.
  - Resource estimates for FTQC
  - Experimental progress
  - Resource estimation
- Technical talk on classical and quantum error correction.



Classical error correction setting

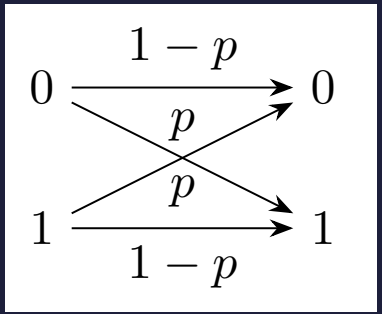


Figure: Bitflip (Binary symmetric channel)

- Information encoded in bits:

  $0 \oplus 0 = 0, \quad 1 \oplus 0 = 0 \oplus 1 = 1, \quad 1 \oplus 1 = 0$

- Bitstrings:

  $$1010 \oplus 1110 = 0100$$

- Noisy channel (environment).
- Bits can be corrupted by noise $0 \mapsto 1$
- Challenge: protect information from noise?

Classical coding: repetition

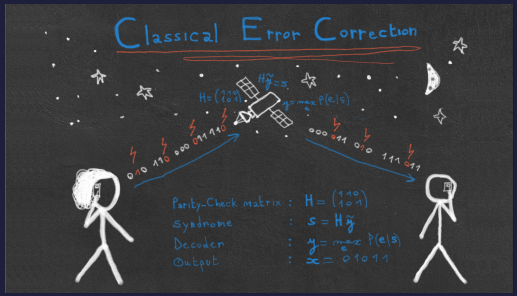


Figure: Sketch of classical error correction

- Encode: $0 \mapsto 000$
- Encode: $1 \mapsto 111$
- Majority vote
- Correct up to one error

Credit: Pesah, All you need to know about classical error correction (2022)

General idea: Error correction

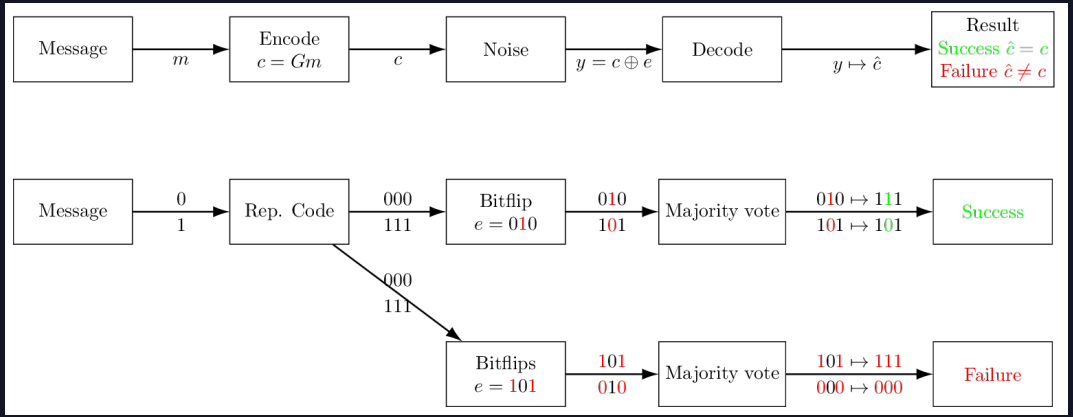


Figure: Error correction

Dissecting the repetition code

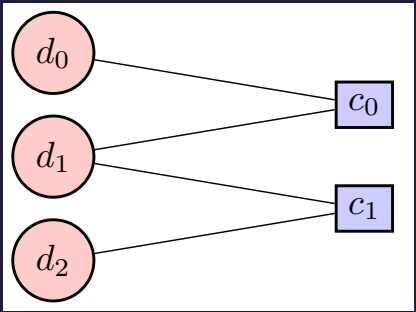


Figure: Tanner graph for 3-bit repetition code

- Parity check matrix: Error detection

$$H = \begin{array}{c} \begin{matrix} d_0 & d_1 & d_2 \end{matrix} \\ \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \end{array} \begin{matrix} c_0 \\ c_1 \end{matrix}$$

  - Check c_0: $d_1 \oplus d_2 = 0$
  - Check c_1: $d_2 \oplus d_3 = 0$

- Syndrome:

$$s = H(c \oplus e) = Hc \oplus He = He$$

- Error $\rightarrow$ nonzero syndrome

More general codes

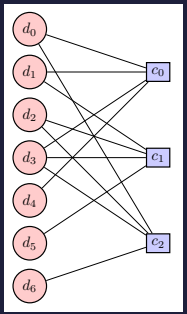


Figure: Tanner graph for the Hamming code

- $[n, k, d]$ code
  - n physical bits
  - k logical bits
  - d distance
  - Correct up to $t = \lfloor (d-1)/2 \rfloor$ bitflips
- Hamming code: $[7, 4, 3]$ code.

$$H = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}$$

More general codes

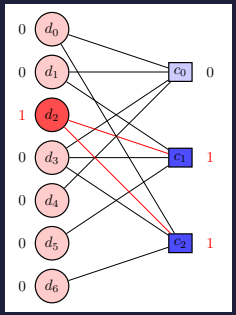


Figure: Tanner graph for the Hamming code

- $[n, k, d]$ code
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$$H = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}$$

Global overview



$$[n, k, d]$$

- k/n encoding rate
- d/n relative distance
- "Good codes": $k/n, d/n = \Omega(1)$
- Easy construction: Random LDPC codes (sparse H)
- Easy decoding: belief propagation

Quantum Information

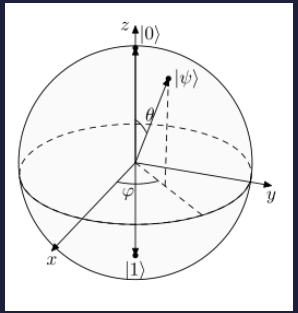


Figure: Bloch sphere representation of a qubit

- Qubit:
$$\alpha |0\rangle + \beta |1\rangle$$
- $\alpha, \beta \in \mathbb{C}$, $|\alpha|^2 + |\beta|^2 = 1$
- More generally
$$|\psi\rangle = \sum_{c \text{ bitstring}} \alpha_c |c\rangle$$
- Entanglement
$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \neq |\psi_1\rangle \otimes |\psi_2\rangle$$

Measurement in Quantum Mechanics

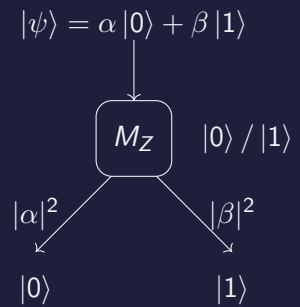


Figure: Meas. in the computational basis

- Measurement: projective, probabilistic.
- In arbitrary basis (importance of phase!):

$$|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$$

Measurement in Quantum Mechanics

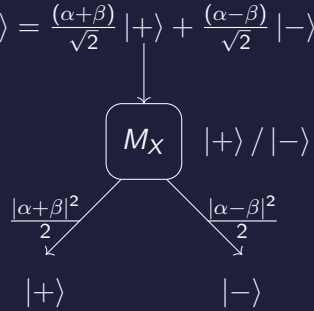


Figure: Meas. in the X basis

- Measurement: projective, probabilistic.
- In arbitrary basis (importance of phase!):

$$|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$$

No-cloning theorem



Statement

- Impossible to copy an arbitrary quantum state $|\psi\rangle$.
- No unitary U such that

$$\forall\, |\psi\rangle\,, \quad U\,|\psi\rangle \otimes |0\rangle = |\psi\rangle \otimes |\psi\rangle$$

Implications

- Cannot copy quantum information
- Information when measured is lost
- Challenge for measuring syndromes.

Wootters & Zurek, Nature 299, 802–803 (1982)

Quantum errors: two axes

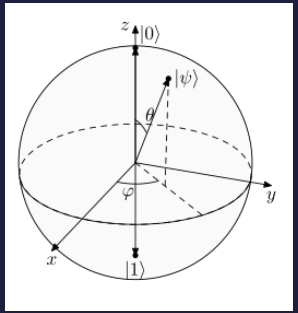


Figure: Bloch sphere representation of a qubit

- Bit flip: X:

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle,$$

- Phase flip: Z

$$Z|+\rangle = |-\rangle, \quad Z|-\rangle = |+\rangle$$

- Combined: $Y = iXZ$
- Continuous errors: $U = u_I I + u_X X + u_Y Y + u_Z Z$
- Error correction should protect against bit and phase flips.

Challenges for QEC

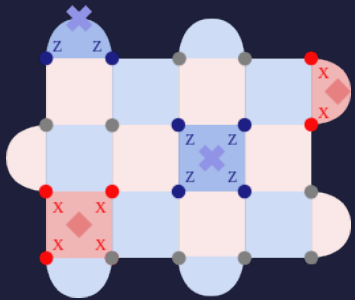


Figure: The Surface code

- Protect against bit and phase flips
- Finding codes: much harder than for classical.
- Checks are now multi-qubit Pauli operators:

$$K\,|\psi_L\rangle = |\psi_L\rangle$$

with $|\psi_L\rangle$ any quantum states

- Simplest example: the surface code
- $[\![n = d^2, k = 1, d]\!]$ code.

QEC codes: Take Home Messages

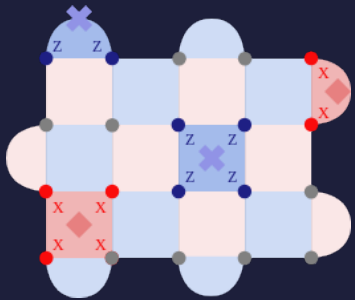


Figure: The Surface code

- Bit and phase flip error protection
- Restrictions on how to processs quantum information
- More challenging than classical codes
- Existing code constructions: CSS codes
  - two classical codes H_Z and H_X parity check matrices
  - H_Z for bit flips
  - H_X for phase flips
  - CSS conditions: $H_Z H_X^T = 0$ (commuting checks)

What do you need to build a FT quantum computer?

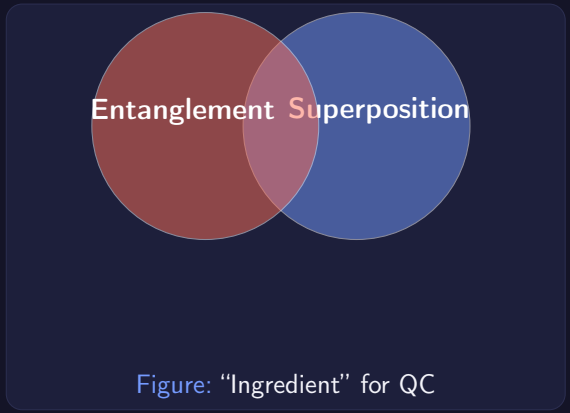


Figure: "Ingredient" for QC

- Low noise qubits / operations
- Perform logic
- Universal set of gates
- Clifford gates:
  - X bitflip: $X|0/1\rangle = |1/0\rangle$
  - Z phase-flip: $Z|\pm\rangle = \pm|\pm\rangle$
  - Hadamard: $H|0\rangle = |+\rangle$
  - Phase: $S|+\rangle = |+i\rangle$
  - CNOT $|+,0\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$

Gottesman-Knill no-go theorem



Statement

- Clifford circuits:
  - $|0\rangle$ state preparation
  - Clifford gates
  - Pauli measurements
- Classically efficient simulation

Implications

- Clifford circuits are not universal
- No quantum advantage of Clifford circuits
- Need non-Clifford resources ("Magic")
- Example $T = \sqrt{S} = \sqrt[4]{Z}$ gate
- Canonical extension: Clifford + T universality

Gottesman, ArXiv 9807006 (1998)

What do you need to build a FT quantum computer?

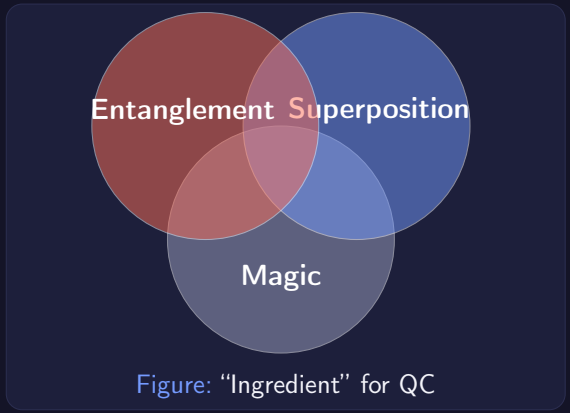


Figure: "Ingredient" for QC

- Low noise qubits / operations
- Perform logic
- Universal set of gates
- Clifford gates
- Gottesman-Knill no-go theorem.
- Clifford + T Universal

Eastin-Knill no-go theorem



Statement

- Transversal gates for a QEC code:

$$U_L = \prod_i U_i$$

- Universal gate sets (e.g. Clifford + T).
- Transversal gate set $\neq$ Universal gate set.

Implications

- Transversal gates: "Easy way" of doing gates
- Need other ways to perform universal gates on logical qubits
- Hard to compute on logical qubits
- Additional resource overhead

Eastin & Knill, Phys. Rev. Lett. 102, 110502 (2009)

Performing gates: Ways around

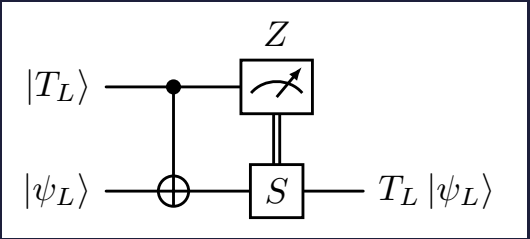


Figure: Magic state injection.

- Magic state distillation
- Lattice surgery
- Code switching
- Lead to additional resource overhead.

Performing gates: Ways around

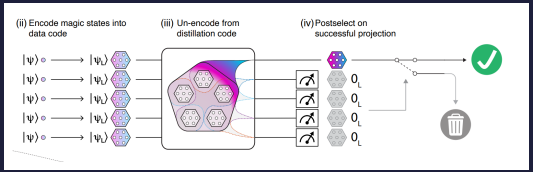


Figure: Magic state distillation.

- Magic state distillation
- Lattice surgery
- Code switching
- Lead to additional resource overhead.

Rodriguez et al., Nature 645, 620–625 (2025)

Performing gates: Ways around

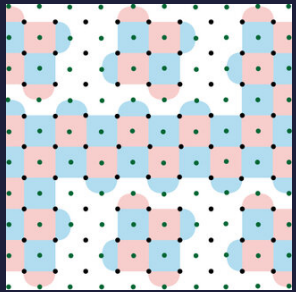


Figure: Lattice surgery.

- Magic state distillation
- Lattice surgery
- Code switching
- Lead to additional resource overhead.

Lin et al., Quantum Sci. Technol., 10, 035007 (2025)

Performing gates: Ways around

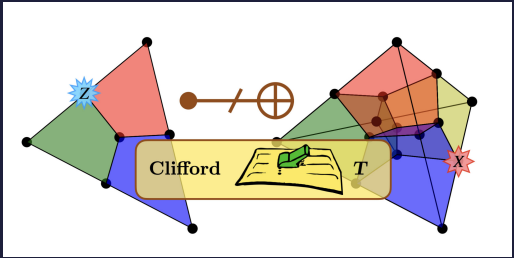


Figure: Code switching.

- Magic state distillation
- Lattice surgery
- Code switching
- Lead to additional resource overhead.

Heußen et al., Quantum 9, 1846 (2025).



Why the surface code?

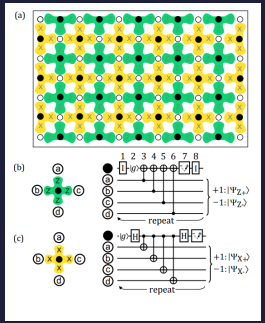


Figure: The Surface code

- Local checks
- 2D nearest-neighbor layout
- Highest circuit-level threshold ($\approx 1\%$)
- Mature decoding tools
- We know how to do FTQC with it

Fowler et al., Phys. Rev. A 86, 032324 (2012)

Code distance costs qubits

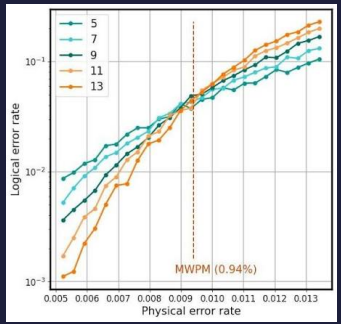


Figure: Threshold of the surface code

- Larger d: lower p_L

$$p_L \propto (p/p_{\mathrm{th}})^{(d+1)/2}$$

- Code block: $O(d^2)$ qubits
- Teraquop regime ($p_L \lesssim 10^{-12}$), $d \approx 30$
- $d^2 \approx 1000$ qubits per logical qubit...
- FTQC leads to huge resource overhead

Caune et al., ArXiv 2306.17142 (2023)

Surface code FTQC

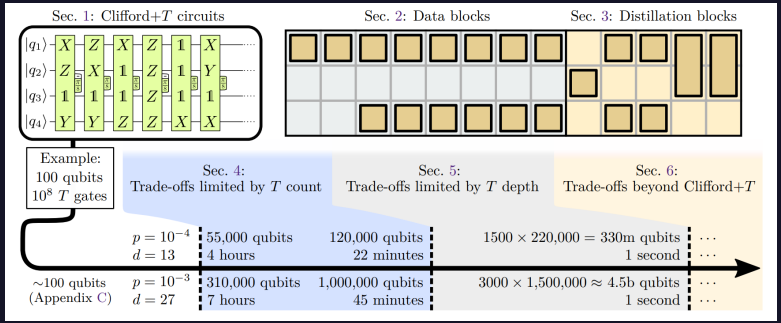


Figure: Surface code quantum computing, Litinski, Quantum 3, 128 (2019)

- Pauli-product measurements
- 2D lattice
- *T*-count limited
- Magic state distillation overheads

FTQC with surface code

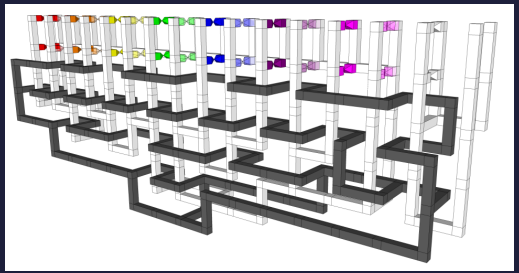


Figure: Space-time footprint of surface code QC

- Fault-tolerant quantum computing is possible
- Compatible with planar connectivity
- Huge overheads...
  - k logical qubits to $n \gg k$ physical qubits
  - FT quantum processing
  - Magic-state factories
- Resource estimation of FTQC!

Fowler, Devitt, ArXiv 1209.0510 (2012)



Why resource estimation?

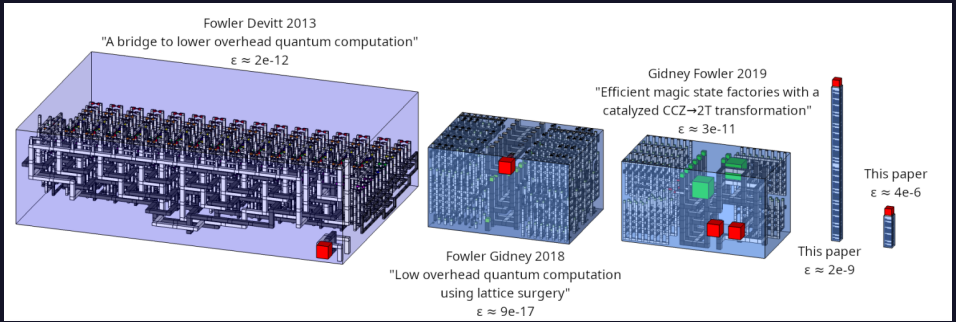


Figure: Architecture improvements, Gidney et al., ArXiv 2409.17595 (2024)

- Resources: qubits, time
- Anticipate bottlenecks and trade-offs
- Motivate experimental developments
- Baseline for architecture improvements

Resource estimation assumptions: Objectives

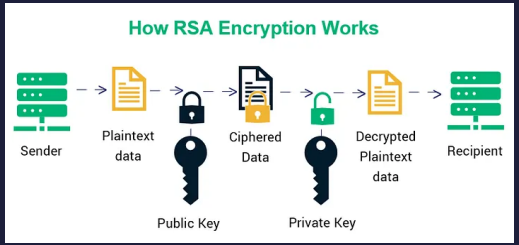


Figure: RSA-2048

- Useful
- Industry relevant
- Complexity theory garantees
- A quantum circuit implementation
- Cryptographically relevant problems (RSA-2028, ECC-256)

Credit Krishna Kaiwartya, Medium 2025

Resource estimation assumptions: Other assumptions

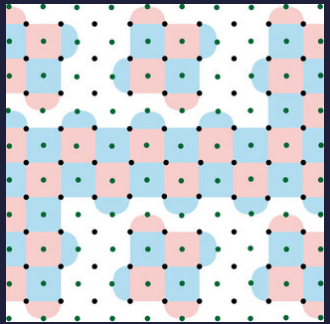


Figure: 2D layout

- Qubit connectivity
  - Arbitrary connectivity
  - 2D nearest-neighbor grid
- Clock-cycle time and reaction time
  - Hardware-dependent
  - Decoder / code dependent
- FT Gate implementations

Example: Gidney, Ekerå(2021)

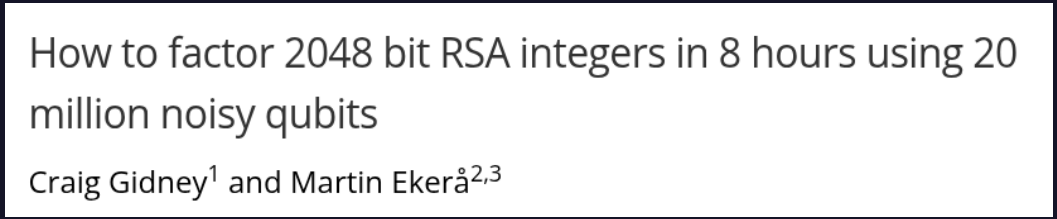


Figure: Gidney, Ekerå, Quantum 5, 433 (2021)

Example: Gidney, Ekerå(2021)

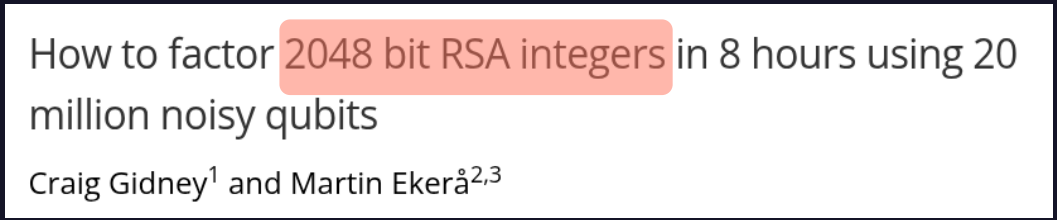


Figure: Gidney, Ekerå, Quantum 5, 433 (2021)

Example: Gidney, Ekerå(2021)

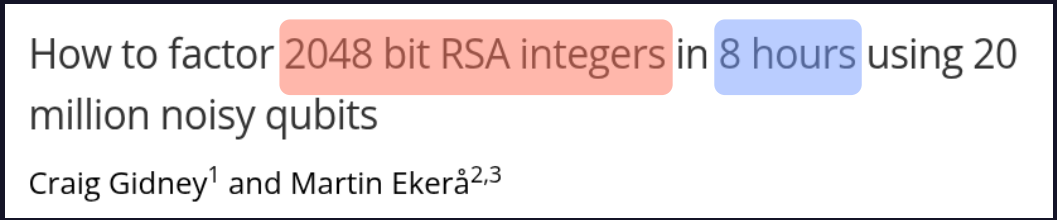


Figure: Gidney, Ekerå, Quantum 5, 433 (2021)

Example: Gidney, Ekerå(2021)

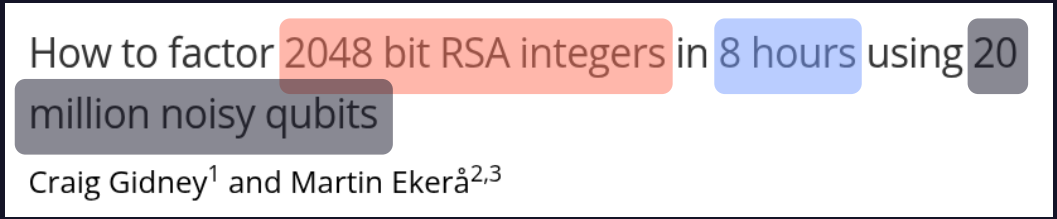


Figure: Gidney, Ekerå, Quantum 5, 433 (2021)

Example: Gidney, Ekerå(2021)

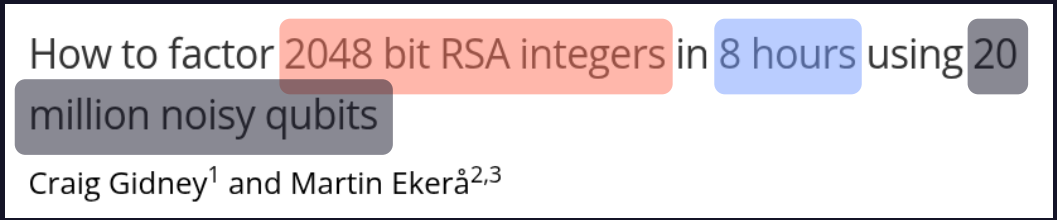


Figure: Gidney, Ekerå, Quantum 5, 433 (2021)

- 2D planar connectivity
- 10^{-3} gate error
- 1 μs cycle; 10 μs reaction
- Surface code
- Lattice surgery
- Magic state distillation

Example: Gidney, Ekerå(2021)

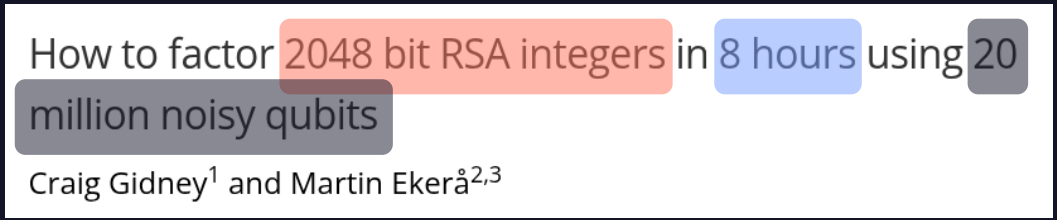


Figure: Gidney, Ekerå, Quantum 5, 433 (2021)

- Full-stack simulation
- Single-chip architecture
- Resource overhead: is it realistic?

- A landmark point
- Benchmark improvements
- Architecture co-design

Example: Gidney, Ekerå(2021)

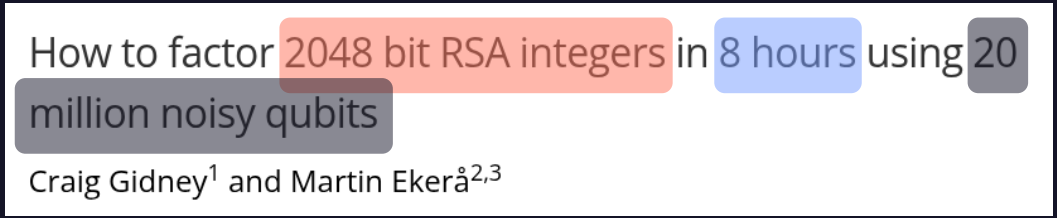


Figure: Gidney, Ekerå, Quantum 5, 433 (2021)

Assumptions

- Realistic?
- Correct?
- What can be improved?
- Surface code really the best?
- Noise model?
- 2D grid?

Can we improve on it?

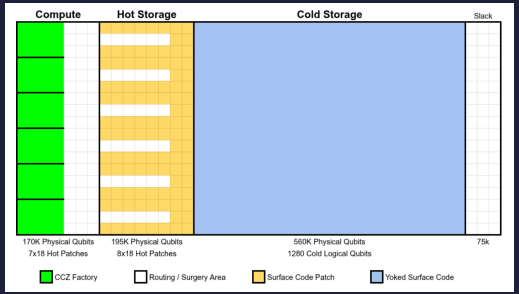


Figure: Architecture improvements

- Same assumptions
- Improvements on quantum algorithms
- Other code: "Yoked surface code" for storage
- Magic-state cultivation over distillation
- $< 1M$ qubits, < 1 week
- Guidance for experimental realizations
- Can we do much better?

Gidney, Arxiv 2505.15917 (2025)

Can we improve on it?

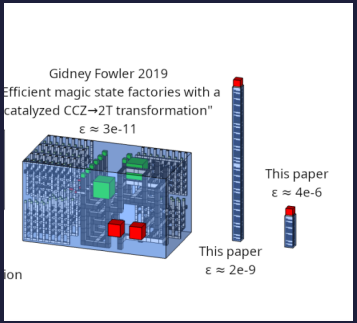


Figure: Magic state cultivation

- Same assumptions
- Improvements on quantum algorithms
- Other code: “Yoked surface code” for storage
- Magic-state cultivation over distillation
- $< 1M$ qubits, < 1 week
- Guidance for experimental realizations
- Can we do much better?

Gidney, Arxiv 2505.15917 (2025)

Can we improve on it?



Figure: Resource estimate (qubit count over the years)

- Same assumptions
- Improvements on quantum algorithms
- Other code: "Yoked surface code" for storage
- Magic-state cultivation over distillation
- $< 1M$ qubits, < 1 week
- Guidance for experimental realizations
- Can we do much better?

Gidney, Arxiv 2505.15917 (2025)



Experimental realizations

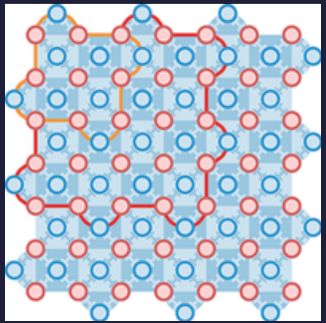


Figure: Surface code

- Clear path from resource estimations
- Target error rate
- Quantum error correction demonstration
- Logical gate primitives.

He et al., Phys. Rev. Lett. 135, 260601 (2025)

From break-even to below-threshold QEC

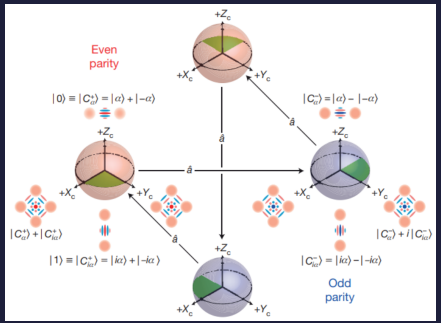


Figure: Cat qubit code beyond break-even

- Break-even demonstration $p_L < p$
- Below-threshold operation
- On different hardware platforms:
  - Superconducting qubits, $\Lambda \approx 2$, real-time decoding
  - Trapped ions, $\Lambda \approx 12$

Ofek et al., Nature 536, 441–445 (2016)

From break-even to below-threshold QEC

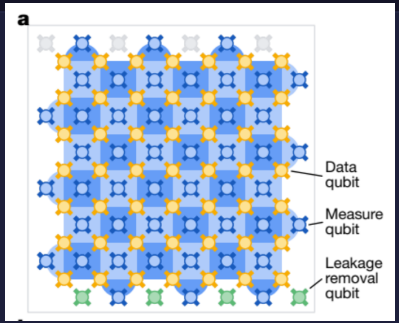


Figure: Below-threshold QEC with surface codes and superconducting platforms

- Break-even demonstration $p_L < p$
- Below-threshold operation
- On different hardware platforms:
  - Superconducting qubits, $\Lambda \approx 2$, real-time decoding
  - Trapped ions, $\Lambda \approx 12$

Google Quantum AI, Nature 638, 920–926 (2025)

From break-even to below-threshold QEC

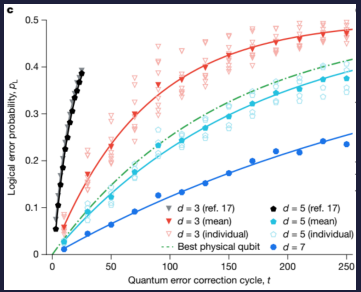


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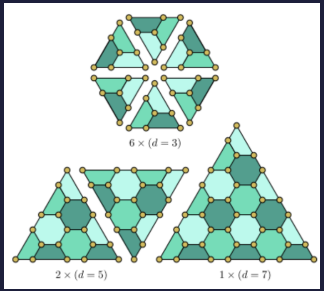


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- Below-threshold operation
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  - Trapped ions, $\Lambda \approx 12$

Lavasani et al. (2026)

Experimental logical gates

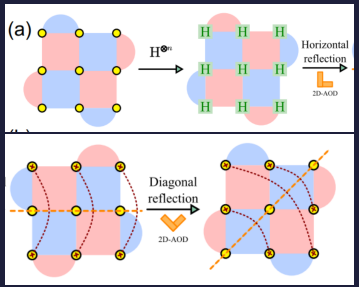


Figure: Transversal gates on neutral-atom qubits

- Logical memories
- Demonstrations of logical gates
  - Transversal gates
  - Lattice surgery
  - Magic state distillation
  - Magic state cultivation
  - Code switching

Pogorelov et al., Nature Physics 21, 298–303 (2025)

Experimental logical gates



Figure: Lattice surgery with trapped ions

- Logical memories
- Demonstrations of logical gates
  - Transversal gates
  - Lattice surgery
  - Magic state distillation
  - Magic state cultivation
  - Code switching

Schindler et al., Nature 589, 220–224 (2021)

Experimental logical gates

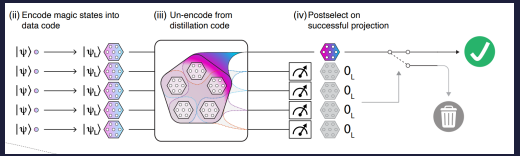


Figure: Magic-state distillation with cold atoms

- Logical memories
- Demonstrations of logical gates
  - Transversal gates
  - Lattice surgery
  - Magic state distillation
  - Magic state cultivation
  - Code switching

Sales Rodriguez et al., Nature 645, 620–625 (2025)

Experimental logical gates

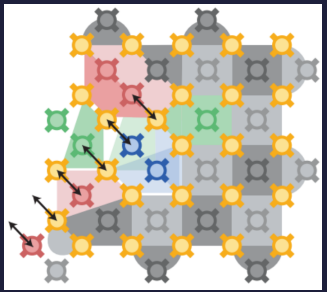


Figure: Magic-state cultivation

- Logical memories
- Demonstrations of logical gates
  - Transversal gates
  - Lattice surgery
  - Magic state distillation
  - Magic state cultivation
  - Code switching

Rosenfeld et al., Science (2026)

Experimental logical gates

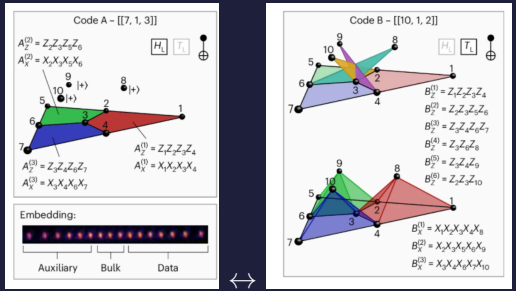


Figure: $[\![7,1,3]\!] \leftrightarrow [\![10,1,2]\!]$ Code switching

- Logical memories
- Demonstrations of logical gates
  - Transversal gates
  - Lattice surgery
  - Magic state distillation
  - Magic state cultivation
  - Code switching

Rosenfeld et al., Science (2026)

Entering the “Early-FTQC” era

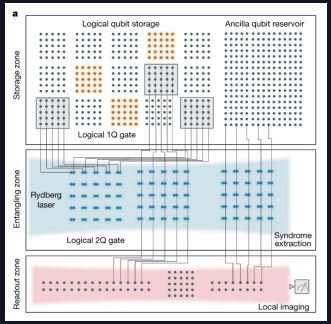


Figure: Early FTQC platforms, Bluvstein et al., Nature 626, 58–65 (2024)

- 48 logical qubits ($[\![8,3,2]\!]$)
- transversal CZ, CCZ gates . . .
- Milestone demonstrations
- not yet application scale



Theoretical advances in QEC

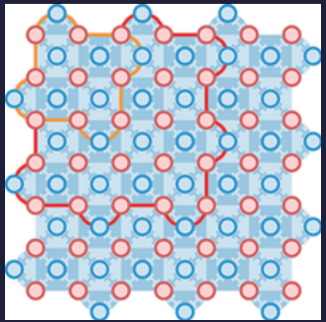


Figure: Surface code

- Goal reducing FTQC overheads
- Ex: Magic-state distillation $\rightarrow$ Cultivation
- Surface code $[\![d^2, 1, d]\!]$ scaling issue...

$$\frac{k}{n} = \frac{1}{d^2} \xrightarrow[d\to\infty]{} 0$$

- Reducing errors $\implies$ reducing the rate
- Classical LDPC codes $[n, \Theta(n), \Theta(n)]$
- Existence of good qLDPC codes?

He et al., Phys. Rev. Lett. 135, 260601 (2025)

The Quest for "Good" qLDPC codes

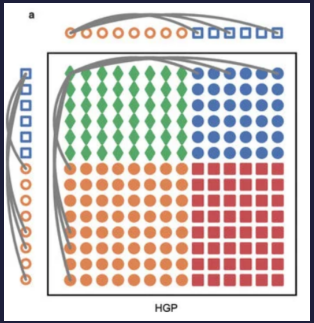


Figure: Hypergraph product codes

Tillich & Zémor, IEEE Transactions on Information Theory, 60, 2 (2014)

- 2013: Hypergraph product codes $[\![n, \Theta(n), \Theta(n^{\frac{1}{2}})]\!]$
- 2021: Balanced-product codes $[\![n, \Theta(n^{\frac{4}{5}}), \Omega(n^{\frac{3}{5}})]\!]$
- 2021-2022: Lifted-product codes $[\![n, \Theta(n), \Theta(n)]\!]$
- 2022: Quantum Tanner codes $[\![n, \Theta(n), \Theta(n)]\!]$
- Assymptotically good qLDPC codes exists. Explicit constructions?

Panteleev & Kalachev, Arxiv 2111.03654 (2021)

Other efficient qLDPC code constructions

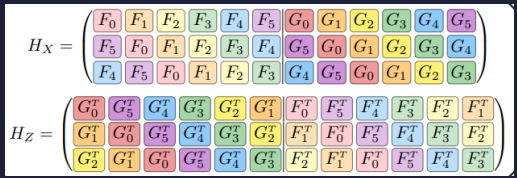


Figure: Affine Permutation Matrix (APM) codes

- Bivariate Bycicle codes:
  - Ex:$[\![144, 12, 12]\!]$
  - High threshold code family
- Recent APM codes
  - Ultra-high encoding rates
  - $k/n \geq 1/2$
  - Ex: $[\![9216, 4612, \leq 48]\!]$
- Excellent memories, how to do gates?

Kasai, Quantum 10, 2205 (2026), Zhao et al., Arxiv 2604.16209 (2026)

Promise vs Blocker of qLDPC codes

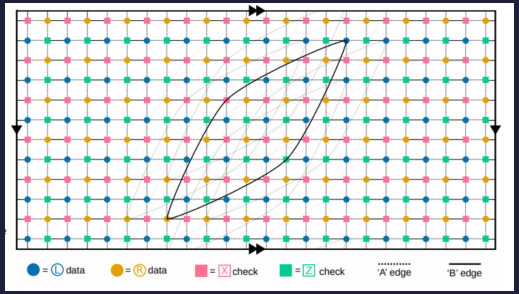


Figure: Bivariate Bycicle codes ($[\![144, 12, 12]\!]$ Gross code)

- Promise
  - Rate
  - Asymptotic scaling
- Blocker
  - Decoding complexity
  - Gate toolbox
  - Hardware mapping
- Challenge: Integrate in FTQC architectures

Bravyi, Nature 627, 778-782 (2024)

Beyond lattice surgery

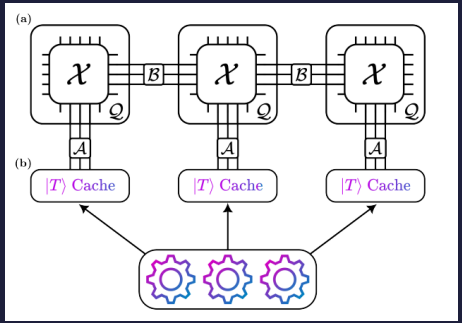


Figure: Extractor architecture for qLDPC codes

He et al., Arxiv 2503.10390 (2025)

- "qLDPC" surgery challenge
  - $k \geq 1$ logical qubits per block
  - Arbitrary Pauli measurements?
- Generalizations of lattice surgery
  - 2022: HGP Surgery,
    Cohen et al., Science Advances, 8, 20 (2022)
  - 2023: CSS/qLDPC surgery
    Cowtan & Burton, Quantum 8, 1344 (2024)
  - 2026: Parallel surgery
    Cowtan et al., PRX Quantum 7, 020325 (2026)
  - 2026: Clifford-measurement surgery
    Chang et al., ArXiv 2610.03920 (2026)

Recent resource estimations

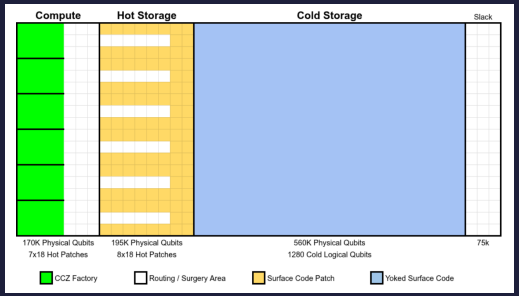


Figure: Superconducting architecture

- Gidney et al., (2025) for superconducting qubits
- Assumptions:
  - 2D connectivity
  - Surface code
  - 1 μs cycle
  - Lattice surgery

Gidney, Arxiv 2505.15917 (2025)

Recent resource estimations

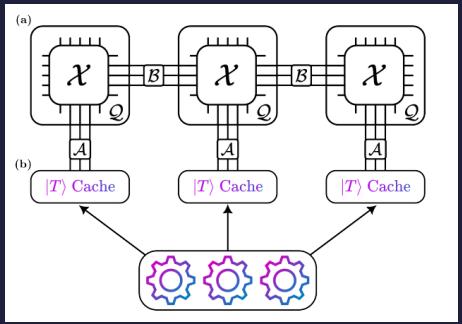


Figure: Modular qLDPC architecture

- Atomic platforms
- New Assumptions:
  - 2D connectivity
  - Surface code
  - 1 μs cycle
  - Lattice surgery

He et al., Arxiv 2503.10390 (2025)

Recent resource estimations

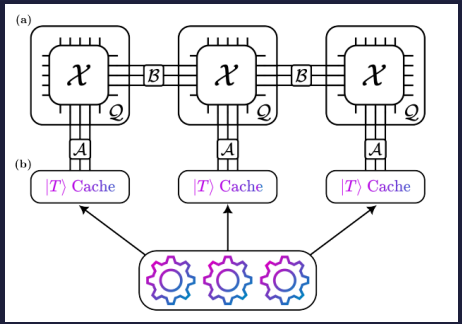


Figure: Modular qLDPC architecture

- Atomic platforms
- New Assumptions:
  - ~~2D connectivity~~, Arbitrary connectivity
  - ~~Surface code~~, qLDPC codes
  - ~~1 μs cycle~~, 1 ms cycle
  - ~~Lattice surgery~~, qLDPC surgery

He et al., Arxiv 2503.10390 (2025)

The Pinnacle Architecture

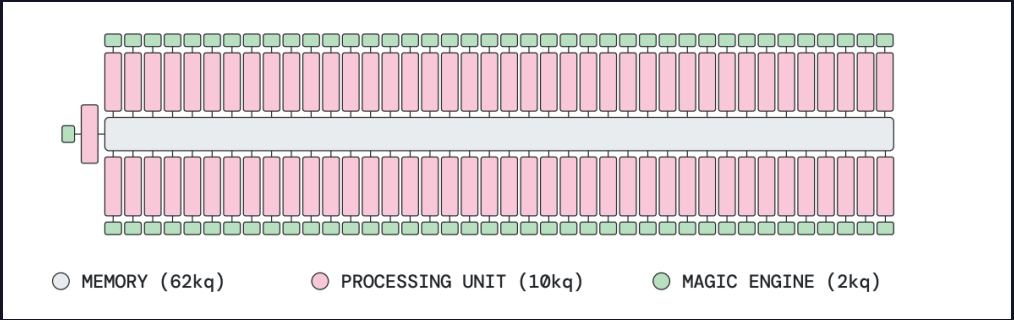


Figure: The Pinnacle architecture for qLDPC codes, Webster et al., ArXiv 2602.11457 (2026)

- Modular architecture
- 3 code families
- Less than 100k physical qubits
- Space-time trade-offs

The Pinnacle Architecture

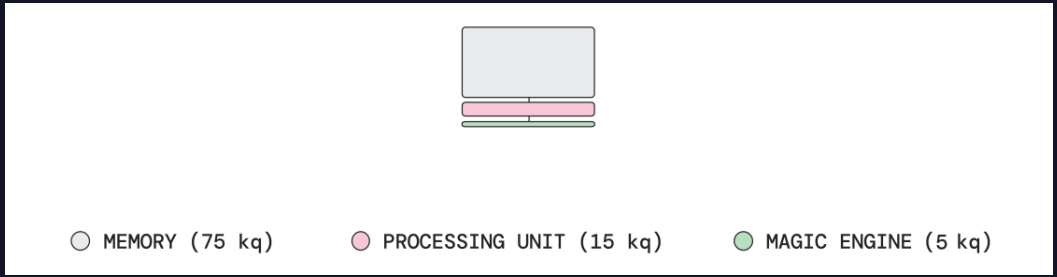


Figure: The Pinnacle architecture for qLDPC codes, Webster et al., ArXiv 2602.11457 (2026)

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Oratomic's architecture

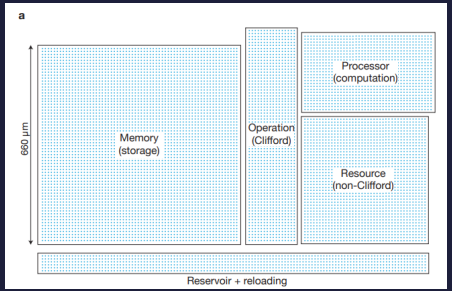


Figure: Oratomic's architecture for qLDPC codes

- Break RSA-2048
- qLDPC codes (LP) ($[\![278, 1480, \leq 24]\!]$)
- With less than 10k physical qubits!
- . . . in >100 years
- Space overhead seems feasible, but time overhead is huge
- Are these assumptions realistic?

Cain et al., ArXiv 2603.28627 (2026)

Oratomic's architecture

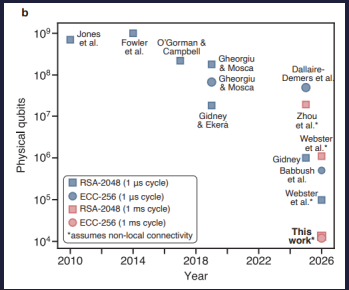


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Other (more realistic) architectures

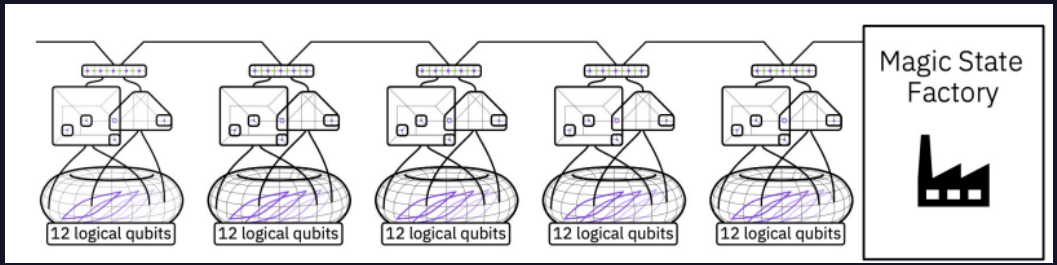


Figure: IBM's architecture with $[\![288, 12, 18]\!]$ codes

Other (more realistic) architectures

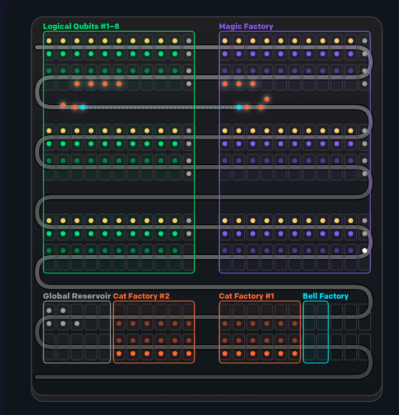


Figure: Ion Q's walking cat architecture with $[\![90, 8, 10]\!]$ codes

Tripier et al., ArXiv 2604.19481 (2026)



Main takeaways

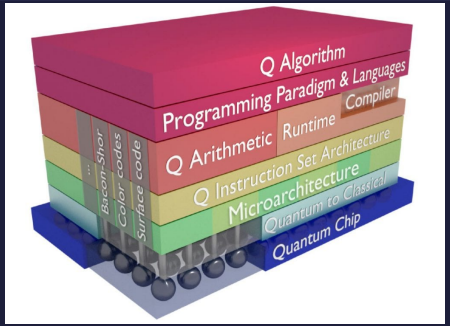


Figure: Stacks of a FT Quantum Computer

- NISQ and FTQC are very different
- QEC leads to huge overhead
- No-go theorems make it non-trivial
- QEC is now an experimental reality
- Many approaches are being explored
- FTQC resource estimations

credit: QuTech

Closing remarks



What is now established

- Logical error suppression: demonstrated
- Core FT primitives: demonstrated
- FTQC engineering: no longer hypothetical

What is still hard

- Full-stack overheads: still large
- Non-Clifford resources: dominant bottleneck
- Decoder-control-hardware integration

Why am I optimistic about FTQC?

- Multiple pathways progressing in parallel
- Resource estimates improving quickly
- But what for?

FTQC: No longer a question of principle.
It is a system engineering problem.

Thank you for your attention – Questions are welcome